0. **Code:** vemuri 6191.zip

1. vemuriu 6191 / Array / Program.cs
2. vemuriu 6191 / dictionary / Program.cs
3. vemuriu 6191 / loop / Program.cs

1. vemuriu 6191 / Array / Program.cs

```
namespace Array
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int[] testArray = {
                23,
                1,
                10,
                5,
                2
            };
            int[] insertionCounts = GetInsertionCounts(testArray);
            Console.WriteLine("Insertion counts fo each element:");
            for (int i = 0; i < insertionCounts.Length; i++)
            {
                Console.WriteLine(insertionCounts[i]);
            }
        }
        static int[] GetInsertionCounts(int[] array)
        {
            int[] counts = new int[array.Length];
            for (int i = 0; i < array.Length; i++)
            {
                int insertionCount = 0;
                for (int j = 0; j < i; j++)
                {
                    if (array[j] > array[i])
                    {
                        insertionCount++;
```

B1

B2

j < i

B3

1. **vemuriu 6191 / Array / Program.cs**

```
                    }
                }
                counts[i] = insertionCount;
            }
            return counts;
        }
    }
}
```

B3

134

without sorting

2. vemuriu 6191 / dictionary / Program.cs

```
using System.Diagnostics.CodeAnalysis;

namespace dictionary
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int[] inputArray = {
                1,
                2,
                3,
                2,
                7,
                3,
                4,
                4,
                5,
                5,
                4,
                5,
                1,
                6,
                1,
                1
            };
            int result = SumOfMaxNumbersByFrequency(inputArray);
            Console.WriteLine("Sum of all largest numbers by frequency:"
              + result);
        }
        static int SumOfMaxNumbersByFrequency(int[] array)
```

2. **vemuriu 6191 / dictionary / Program.cs**

```
        {
            if (array == null)
                return 0;
            Dictionary < int, int > occurrences = new Dictionary < int,
              int > ();
            Dictionary < int, int > maxByFrequency = new Dictionary <
              int, int > ();

            for (int i = 0; i < array.Length; i++)
            {
                int number = array[i];
                if (occurrences.ContainsKey(number))
                    occurrences[number]++;
                else
                    occurrences[number] = 1;
            }
            var keys = new List < int > (occurrences.Keys);
            for (int i = 0; i < keys.Count; i++)
            {
                int number = keys[i];
                int frequency = occurrences[number];
                if (!maxByFrequency.ContainsKey(frequency) || number >
                  maxByFrequency[frequency])
                    maxByFrequency[frequency] = number;
            }
            int sum = 0;
            foreach(KeyValuePair < int, int > pair in maxByFrequency)
            {
                sum += pair.Key * pair.Value;
            }
            return sum;
```

C5

C2

Use: foreach(var kvp in occurrences)

= kvp.Value;

= kvp.Key;

C3

C4

2. **vemuriu 6191 / dictionary / Program.cs**

```
        }
    }
}
```

Looks good.

3. vemuriu 6191 / loop / Program.cs

```
namespace test3

{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("Enter an integer:");
            int number = Convert.ToInt32(Console.ReadLine());
            int baseN = 0;
            while (baseN ≤ 0)
            {
                Console.WriteLine("Enter base-N(must be positive):");
                baseN = Convert.ToInt32(Console.ReadLine());

            }
            int sumOfDigits = 0;
            int maxDigit = 0;
            int tempnumber = number ≥ 0 ? number : -number;

            while (tempnumber > 0)
            {
                int digit = tempnumber % baseN;
                sumOfDigits += digit;
                if (digit > maxDigit)
                {
                    maxDigit = digit;
                }
                tempnumber /= baseN;
            }
```

A1 A2 A5 A3 A3 A4 A5 A3

3. **vemuriu 6191 / loop / Program.cs**

```
            Console.WriteLine($"Sum of base-{baseN}digits: {sumOfDigits}.");
            Console.WriteLine($"Max of base-{baseN}digit: {maxDigit}.");
        }

    }
}
```